

# PROGRAMMING SUMMARY

## DESCRIPTION

We programmed a few modules:

- A 128x64 OLED display to show the level of soap as a percentage
- Light sensor/Photoresistor to activate the pump when a hand is blocking light
- Laser Sensor measure the amount of soap in the container
- The pump and a Mosfet. The pump is activated based on the reading of the photoresistor
- The microcontrollers itself(ESP32&Arduino) to control all of these module and to implement the messaging capabilities.

## IMPORTANT DETAILS

- We divide the tasks between the arduino and esp32 to spread the load
- The OLED display uses the measurement from the laser, we map that value and using a few libraries, we display the percentage.
- 2 of our components use the I2c communication protocol. Since the esp32 only has one I2c bus(GPIO 22&21), both components have to be connected in parallel to the bus and MUST be supplied the same voltage
- The Mosfet is used to turn the pump on and off because we cannot do that by connecting the arduino pins directly, it's pins can't handle the current drawn[13]
- The esp32 has wifi capabilities, which we use to send messages using a whatsapp bot. When this feature is used, ADC\_2 pins go offline.
- The photoresistor isn't useful in the dark, there must be a light source. To resolve this, we shine a bright LED close to the photoresistor when in the dark to trick it into thinking it's bright.

[Github Code](#)

Reference:

[13] The Engineering Mindset. (2024, February 19). *MOSFET Explained - How MOSFET works* [Video]. YouTube. [https://www.youtube.com/watch?v=AwRlSze\\_9m4](https://www.youtube.com/watch?v=AwRlSze_9m4)







